Machine Learning Exam Preparation – 40 Important Questions

Fundamentals

1. Define Machine Learning and explain how it differs from traditional programming with an example.
2. Differentiate between Supervised, Unsupervised, and Reinforcement Learning with one application each.
3. Explain the complete Machine Learning workflow from data collection to deployment.
4. What is data preprocessing? Explain its importance in machine learning projects.
5. Differentiate between feature, label, and dataset with a practical example.

Data Preparation & Preprocessing

6. Explain methods for handling missing values in numerical and categorical datasets.
7. Differentiate between Label Encoding and One-Hot Encoding with examples.
8. What is feature scaling? Compare Min-Max Normalization and Standardization.
9. Explain the concept of imbalanced datasets. How can under-sampling and over-sampling solve this issue?
10. Differentiate between feature selection and feature engineering.

Model Evaluation

11. Why is a dataset split into training and testing sets? What is the purpose of validation data?
12. What is a confusion matrix? Explain TP, TN, FP, and FN.
13. Define Accuracy, Precision, Recall, and F1-score with formulas.
14. Why can accuracy be misleading for imbalanced datasets? Give an example.
15. Explain k-Fold Cross Validation and its advantages over train-test split.

Regression

16. Define Linear Regression. Explain its mathematical equation and assumptions.
17. Compare Mean Absolute Error (MAE), Mean Squared Error (MSE), and RMSE.
18. Why are outliers problematic in Linear Regression?
19. Explain the bias-variance tradeoff in regression models.
20. Why is Linear Regression not suitable for classification problems?

Classification Algorithms

21. Explain Logistic Regression and the role of the Sigmoid function.
22. What is Naive Bayes? Why is it called “naive”?
23. Explain Laplace Smoothing in Naive Bayes.
24. Describe the working of K-Nearest Neighbors (KNN) classifier.
25. How does changing the value of K affect KNN performance?

Decision Trees & Ensemble Learning

26. Explain the structure of a Decision Tree with root node, internal node, and leaf node.
27. Compare Entropy and Gini Impurity in Decision Trees.
28. Why do Decision Trees overfit easily? How can pruning help?
29. What is Random Forest? How does it improve over a single Decision Tree?
30. Differentiate between bagging and boosting.

Support Vector Machines

31. Define Support Vector Machine (SVM). Explain hyperplane, support vectors, and margin.
32. Differentiate between Hard Margin and Soft Margin SVM.
33. What is the Kernel Trick in SVM? Why is it useful?

Clustering

34. What is clustering? Differentiate clustering from classification.
35. Explain the K-Means clustering algorithm step by step.
36. How does the Elbow Method help determine optimal K in K-Means?
37. Compare K-Means and Hierarchical Clustering.
38. Explain Silhouette Score and how it evaluates clustering quality.

Neural Networks & Deep Learning

39. Differentiate between Single Layer Perceptron, Multi-Layer Perceptron, and Deep Neural Networks.
40. Explain activation functions (Sigmoid, Tanh, ReLU) and why they are needed in neural networks.